

MRI Install Checklist

OEM ID	_____	Van ID	_____
Site Name	_____	City, State	_____
Site Contact	_____	Phone Number	_____
SMR Rep	_____	Inspection Date	_____

System Physical Installation

- ☐ Move Cabinets into place in the equipment room, check for best placement and service space
 - PDU closest to main disconnect, If possible
 - RF and Gradient cabinets need room to pull out componets
- ☐ Install pen panel, make sure to fix helium gas lines and coldhead power cables through the pen panel
- ☐ Start installing cables by separating them out into magnet room (blue), console (Green or brown) Equipment (the rest)
- ☐ Make sure to leave a 2ft service loop for the RF cabinet SSM module connections, or any component that pulls out
- ☐ Use cable map to check off all connections and make sure you have all cables
- ☐ Layout console to equipment room cables to make sure of length before running (green or brown is console side)
- ☐ Make sure to make a list of any cables that are needed or missing parts
- ☐ Install MRU on wall, hook up to 120V power or outlet, run MRU cable
- ☐ Make sure you have the Last Best Know shim currents and center frequency of the magnet Magnet Serial #
- ☐ Make a list of anything that is missing or needs to be completed as you are working and double check as you finish
- ☐ Verify that cryogen are ordered and ready for when ready to Ramp and shim the magnet
- ☐ Record Magnet Readings He Level 1 _____ Pressure _____ Flow _____
- ☐ Record Compressor Readings Compressor One -Hrs _____ DY _____ - ST _____ = _____
- ☐ Record Magnet Status Inner/RuO _____ Outer/Coldhead _____ Magnet RuO _____ Magnet Si410 _____ Water Temp _____ Water flow _____

Power Up

- ☐ Verify all power and ground connections to each cabinet ☐ Verify all cable connections to all equipment and magnet
- ☐ Make sure all breakers are turned off to equipment and the main
- ☐ Check main input power voltage with DVM (leg to leg 480V (+or- 10V) and each leg to ground 240 (+or- 10V))
- ☐ As long as main input power is good you can turn on the main breaker, re-check power from cabinet
- ☐ Turn on gradient cabinet fan first to make sure door sucks closed, if door blows open turn power off and swap 2 input phases
- ☐ If E-stop doesn't reset w/bypass then the phase may need to be changed back and phase swapped on gradients (ACGD cabinet)
- ☐ Make sure that system comes all the way up, check table movement, height and home sensor flag
- ☐ Run head and body noise scans, check standard deviation and means for receive signal using an ROI under the image viewer
 - Head noise scan Standard Deviation Mean value
 - Body noise scan Standard Deviation Mean value

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Calibrations

- ☐ Before ramping magnet make sure to de-ice and that shim lead connection is very clean
- ☐ Make sure you have the last best known shim currents and center frequency
- ☐ Follow the ramp procedure for the type of magnet you are working on in the magnet manual
- ☐ Ramp up magnet to the last best known center frequency or closest to the Tesla of the magnet, with AX shim currents
- ☐ After Ax currents are in and the ramp probs are out put in T1 and T2 shim Currents
- ☐ Run first image test scan, same as iso center, check for GE logo orientation (should be in the top right corner)
- ☐ Supercon Shim magnet until the Inhomogeneity is under 100hz and coefficients are under 100 as well
- ☐ Run L-coil, DC offsets, Iso center and gradient cals (if twin speed run zoom mode as well)
- ☐ Run grafidy in all three planes X,Y and Z until all are black or green (if twin speed run zoom mode as well)
- ☐ Re-Run L-coil, DC offsets, then re supercon shim until inhomogeneity under 50hz, check for coefficients to pass specs
- ☐ Re-Run L-coil, DC offsets, Iso center and gradient cals (if twin speed run zoom mode as well)
- ☐ Re-Run grafidy in all three planes X,Y and Z until all are green (if twin speed run zoom mode as well)
- ☐ Run gradient shim until your currents don't change any more
- ☐ Run Echo Planner Imaging scans for head and body (EPT)
- ☐ Check multicoil scans using CTL coil, scan CLTTOP and CTLBOT (if 8ch or 16ch system make sure to check all receivers)
- ☐ Setup network information including PACS, worklists and camera

System Final Installation

- ☐ Make a list of all coils on site and check against the quote of coils sold
- ☐ Make sure to test all coils for signal on each channel
- ☐ Clean up and put every thing away the best that you can
- ☐ Test network Connections and camera to make sure they are working
- ☐ Make a list of anything that is missing or needs follow up
- ☐ Run one last multi-coil, body and head scan before leaving the system